Assignment 2

Suggested Solutions

Q1. Compute (4) and (5) in Proposition 1 of Lecture 4.

Solution.

The market value of the firm at time 0 is

$$\begin{aligned} v(V;V_B) &= V + \mathbb{E}\left[\int_0^\tau \kappa\rho P e^{-rt}dt\right] - (1-\alpha)\mathbb{E}[V_\tau e^{-r\tau}] \\ &= V + \kappa\rho P \cdot \mathbb{E}\left[\int_0^\tau e^{-rt}dt\right] - (1-\alpha)V_B\mathbb{E}[e^{-r\tau}] \\ &= V + \kappa\rho P \cdot \mathbb{E}\left[\frac{1}{r}(1-e^{-r\tau})\right] - (1-\alpha)V_B\mathbb{E}[e^{-r\tau}] \\ &= V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\cdot\mathbb{E}[e^{-r\tau}]. \end{aligned}$$

It follows from (A.1) that we have

$$\mathbb{E}[e^{-r\tau}] = \exp\left\{\log(V_B/V)\cdot\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\right\},$$

where $\xi = \frac{r-\delta}{\sigma^2} - \frac{1}{2}$.

Hence, we get

$$\begin{aligned} v(V;V_B) &= V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\exp\left\{\log(V_B/V)\cdot\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\right\} \\ &= V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right](V_B/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi}. \end{aligned}$$

Next, let us consider the total debt value of the firm.

At time 0 the price of a bond with face value 1 and maturity T when the firm asset $V_0 = V$ is given by

$$B(V;V_B,T) = \mathbb{E}\left[e^{-rT}\mathbf{1}_{\{\tau>T\}} + e^{-r\tau}\cdot ce^{-r(T-\tau)}\mathbf{1}_{\{\tau\le T\}} + \int_0^{\tau\wedge T}\rho e^{-rs}ds\right].$$

Hence, the value of total debt at time 0 is

$$\begin{aligned}D(V;V_B) &= P\int_0^\infty me^{-mT}B(V;V_B,T)dT \\ &= P\int_0^\infty me^{-mT}\mathbb{E}\left[e^{-rT}\mathbf{1}_{\{\tau>T\}} + e^{-r\tau}\cdot ce^{-r(T-\tau)}\mathbf{1}_{\{\tau\le T\}} + \int_0^{\tau\wedge T}\rho e^{-rs}ds\right]dT.\end{aligned}$$

The Laplace transform for the bond price, except the recovery part, is given by

$$\begin{aligned}&\int_0^\infty e^{-mT}\mathbb{E}\left[e^{-rT}I\{\tau>T\} + \int_0^{\tau\wedge T}\rho e^{-rs}ds\right]dT \\ &= \mathbb{E}\left\{\int_0^\infty e^{-mT}\left[e^{-rT}I\{\tau>T\} + \frac{\rho}{r}\left(1-e^{-r(\tau\wedge T)}\right)\right]dT\right\} \\ &= \mathbb{E}\left\{\int_0^\tau e^{-mT}e^{-rT}dT + \int_0^\tau e^{-mT}\frac{\rho}{r}\left(1-e^{-rT}\right)dT + \int_\tau^\infty e^{-mT}\frac{\rho}{r}\left(1-e^{-r\tau}\right)dT\right\} \\ &= \mathbb{E}\left\{\left(1-\frac{\rho}{r}\right)\int_0^\tau e^{-(m+r)T}dT + \int_0^\infty e^{-mT}\frac{\rho}{r}dT - \frac{\rho}{r}e^{-r\tau}\int_\tau^\infty e^{-mT}dT\right\} \\ &= \mathbb{E}\left\{\left(1-\frac{\rho}{r}\right)\frac{1-e^{-(m+r)\tau}}{m+r} + \frac{\rho}{r}\frac{1}{m} - \frac{\rho}{r}\frac{e^{-(m+r)\tau}}{m}\right\} \\ &= \frac{\rho+m}{m(r+m)}(1-\mathbb{E}[e^{-(r+m)\tau}]),\end{aligned}$$

which the first equality follows by changing the order the integration. Now, we only need to compute the Laplace transform of the recovery part

$$\begin{aligned}&\int_0^\infty e^{-mT}\mathbb{E}\left[e^{-r\tau}\cdot ce^{-r(T-\tau)}I\{\tau\le T\}\right]dT \\ &= c\int_0^\infty e^{-(m+r)T}\mathbb{E}\left[I\{\tau\le T\}\right]dT \\ &= c\mathbb{E}\left\{\int_0^\infty e^{-(m+r)T}I\{\tau\le T\}dT\right\} \\ &= c\mathbb{E}\left\{\int_\tau^\infty e^{-(m+r)T}dT\right\} \\ &= \frac{c}{m+r}\mathbb{E}[e^{-(m+r)\tau}].\end{aligned}$$

In summary, the Laplace transform of the bond price is given by

$$\begin{aligned}&\int_0^\infty e^{-mT}B(V;V_B,T)dT \\ &= \frac{\rho+m}{m(r+m)}(1-\mathbb{E}[e^{-(r+m)\tau}]) + \frac{c}{m+r}\mathbb{E}[e^{-(m+r)\tau}] \\ &= \frac{\rho+m}{m(r+m)} + \left(\frac{c}{m+r} - \frac{\rho+m}{m(r+m)}\right)\mathbb{E}[e^{-(m+r)\tau}] \\ &= \frac{\rho+m}{m(r+m)} - \left(\frac{\rho}{r}\frac{1}{m} + \left(1-\frac{\rho}{r}-c\right)\frac{1}{r+m}\right)\exp\left\{\log(V_B/V_0)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}(m+r)}+\xi\right)\right\}.\end{aligned}$$

Since $c = \frac{m+r}{m}\frac{\alpha V_B}{P}$, the value of total debt at time 0 is

$$
\begin{aligned}
D(V;V_B) &= P\int_0^{+\infty} me^{-mT}B(V;V_B,T)dT \\
&= P\frac{\rho+m}{r+m} - P\left(\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right)\exp\left\{\log(V_B/V)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}(m+r)}+\xi\right)\right\} \\
&= P\frac{\rho+m}{r+m} - P\left(\frac{\rho}{r} + \left(1-\frac{\rho}{r}-\frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right)(V_B/V)^{\sqrt{\xi^2+2(m+r)/\sigma^2}+\xi}\,.
\end{aligned}
$$

Q2. The total equity value of the firm is given by

$$\begin{aligned}
&S(V;V_B) = v(V;V_B) - D(V;V_B) \\
&= V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right](V_B/V)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi} \\
&\quad - P\frac{\rho+m}{r+m} + P\left[\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right](V_B/V)^{\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi}.
\end{aligned}$$

Show that the solution of

$$\frac{\partial S(V;V_B^*)}{\partial V}|_{V=V_B^*} = 0$$

is given by $V_B^* = \epsilon P$, where

$$\epsilon = \frac{\frac{\rho+m}{r+m}\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right) - \frac{\kappa\rho}{r}\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)}{1+(1-\alpha)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)+\alpha\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)}.$$

Solution.

It follows from $S(V;V_B)$ that

$$\begin{aligned}
&\frac{\partial S(V;V_B)}{\partial V} \\
&= 1 - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\left(\frac{V_B}{V}\right)^{\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi-1}\left(-\frac{V_B}{V^2}\right) \\
&\quad + P\left[\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)\left(\frac{V_B}{V}\right)^{\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi-1}\left(-\frac{V_B}{V^2}\right).
\end{aligned}$$

Hence,

$$\begin{aligned}
&\frac{\partial S(V;V_B)}{\partial V}\Big|_{V=V_B} \\
&= 1 - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\left(-\frac{1}{V_B}\right) \\
&\quad + P\left[\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{r+m}\right]\cdot\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)\left(-\frac{1}{V_B}\right).
\end{aligned}$$

Letting

$$\frac{\partial S(V;V_B)}{\partial V}\Big|_{V=V_B} = 0,$$

we obtain

$$\begin{aligned}
&1 + \frac{\kappa\rho}{r}P\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right)\frac{1}{V_B} + (1-\alpha)\left(\sqrt{\xi^2+\frac{2}{\sigma^2}r}+\xi\right) \\
&- P\left[\frac{\rho}{r} + \left(1-\frac{\rho}{r}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right)\frac{1}{V_B} + \alpha\left(\sqrt{\xi^2+\frac{2(r+m)}{\sigma^2}}+\xi\right) = 0,
\end{aligned}$$

which yields

$$V_B^* = \frac{\left[\frac{\rho}{r} + \left(1 - \frac{\rho}{r}\right)\frac{m}{r+m}\right]\left(\sqrt{\xi^2 + \frac{2(r+m)}{\sigma^2}} + \xi\right) - \frac{\kappa\rho}{r}\left(\sqrt{\xi^2 + \frac{2}{\sigma^2}r} + \xi\right)}{1 + (1-\alpha)\left(\sqrt{\xi^2 + \frac{2}{\sigma^2}r} + \xi\right) + \alpha\left(\sqrt{\xi^2 + \frac{2(r+m)}{\sigma^2}} + \xi\right)} P.$$

So $V_B^* = \epsilon P$.

A Laplace Transforms of First Passage Times

Consider the first passage time

$$\tau_b = \min\{t \geq 0 : \ W_{\mu,\sigma}(t) = b\},$$

where $W_{\mu,\sigma}(t)$ is a Brownian motion with drift μ and volatility σ, i.e. $W_{\mu,\sigma}(t) = \mu t + \sigma W(t)$. Note that $W_{\mu,\sigma}(\tau_b) = b$ on the set $\tau_b < \infty$.

For any $s > 0$, the Laplace transform of τ_b is given by

$$\mathbb{E}[e^{-s\tau_b}] = \begin{cases} \exp\left\{-\dfrac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} - \mu\right)\right\}, & \text{if } b > 0, \\ \exp\left\{\dfrac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} + \mu\right)\right\}, & \text{if } b < 0. \end{cases} \tag{A.1}$$